Fibered Renormalization as Representational Dynamics: An Interacting-Block Attack on the Yang–Mills Mass-Gap Programme

Research dossier—restricted theorems, calibrations, and frontier audit

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Abstract

We refine a representation-aware constructive programme for four-dimensional pure Yang–Mills theory. The central correction is that a renormalization step must not be required to contract the degrees of freedom it intentionally retains. It should transport the retained coarse information and contract only the conditional fiber that it eliminates. A two-plaquette heat-kernel block makes this distinction exact: in the Gaussian tangent model the uncentered internal coordinate has boundary response $1/2$, whereas the centered fiber is independent of the retained total holonomy and has response zero. We prove an exact multiscale covariance decomposition, formulate a conditional fibered clustering theorem, prove an independent-defect connected-cluster bound, and give a finite-product regulator comparison. Compact $U(1)$ computations calibrate these claims. None of these results proves four-dimensional basin entry, continuum existence, or a mass gap; those obligations are isolated explicitly.

Status. The compact bridge and Gaussian statements, the covariance identity, the independent-defect bound, and product-TV comparison are PROVED. The clustering implication is CONDITIONAL on stated estimates. The Clay target remains OPEN.

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1 The target and the representational correction

The Clay problem asks, for every compact simple gauge group G, for a nontrivial quantum Yang–Mills theory on $\mathbb{R}^4$ satisfying axiomatic properties and possessing a mass gap $\Delta > 0$ [1]. In Euclidean language a constructive route must produce coherent gauge-invariant Schwinger functions, verify reflection positivity and the remaining reconstruction hypotheses, and obtain decay strong enough to imply a positive spectral threshold after Osterwalder–Schrader reconstruction [2, 3].

The guiding ontology is not that one coordinate system is the theory. Gauge potentials are local coordinates on relational information; a regulator is a temporary realization; a particle is a spectral interpretation after reconstruction; renormalization changes descriptive scale; and the continuum theory is coherence across compatible limiting representations. On this view the mass gap is an invariant form of infrared spectral rigidity.

The earlier one-step question "does every block variable contract?" is too coarse. An RG transformation separates a block into

$$x \longmapsto (y, r), \qquad y = \pi(x),$$

where y is retained and r is integrated out. Relevant information in y need not contract; indeed its survival is the point of blocking. A defensible local question is instead:

> After fixing the retained gauge-invariant coarse data and quotienting gauge redundancy, does the conditional law on the eliminated fiber forget distant boundary perturbations, apart from a quantitatively controlled defect channel?

2 A two-plaquette heat-kernel block

Let $\mathbb{T} = \mathbb{R}/2\pi\mathbb{Z}$ and let

$$p_t(\theta) = \frac{1}{\sqrt{2\pi t}} \sum_{n\in\mathbb{Z}} \exp\left[-\frac{(\theta + 2\pi n)^2}{2t}\right] = \frac{1}{2\pi} \sum_{k\in\mathbb{Z}} e^{-k^2 t/2} e^{ik\theta}$$

be the $U(1)$ heat kernel in the convention used here. It satisfies the exact convolution semigroup identity $p_t * p_u = p_{t+u}$.

Consider two adjacent plaquette angles q_1, q_2 with density $p_t(q_1)p_t(q_2)$ and retain their total holonomy

$$s = q_1 + q_2 \pmod{2\pi}.$$

The internal coordinate is $q = q_1$ and a centered fiber coordinate is $r = q - s/2$ (locally, with the usual circle ambiguity away from the small-field chart).

Proposition 2.1 (Compact heat-kernel bridge, PROVED)**.** *For every $t > 0$ and $s \in \mathbb{T}$, a regular conditional density of q given s is*

$$\gamma_t^s(q) = \frac{p_t(q)p_t(s-q)}{p_{2t}(s)}. \tag{1}$$

Consequently the centered fiber has density $\widetilde{\gamma}_t^s(r) = \gamma_t^s(r + s/2)$.

Proof. The joint density in coordinates (q, s) is $p_t(q)p_t(s-q)$, because Haar measure and the linear change of circle coordinates have unit Jacobian. Its q-integral is $(p_t * p_t)(s) = p_{2t}(s)$. Division gives (1); translation gives the centered formula. □

This is already an interacting conditional law: conditioning couples the two plaquettes. It is exactly solvable because the heat kernel is closed under convolution, a feature familiar in two-dimensional Yang–Mills constructions [4]. It is a calibration model, not a substitute for the four-dimensional nonabelian block.

3 The retained-mode obstruction and its fiber resolution

The small-field tangent model replaces the circle by $\mathbb{R}$. Let $X_1, X_2 \overset{\text{iid}}{\sim} N(0,t)$, retain $S = X_1 + X_2$, and set $R = X_1 - S/2$.

Theorem 3.1 (Exact Gaussian response split, PROVED)**.** *For each $s \in \mathbb{R}$,*

$$X_1 \mid S = s \sim N(s/2, t/2), \qquad R \mid S = s \sim N(0, t/2).$$

Therefore, for the 1-Wasserstein metric,

$$\frac{W_1(\mathcal{L}(X_1 \mid S = s), \mathcal{L}(X_1 \mid S = s'))}{|s - s'|} = \frac{1}{2}, \qquad \frac{W_1(\mathcal{L}(R \mid S = s), \mathcal{L}(R \mid S = s'))}{|s - s'|} = 0.$$

Proof. Gaussian conditioning gives mean $\operatorname{Cov}(X_1, S)\operatorname{Var}(S)^{-1}s = (t/2t)s = s/2$ and conditional variance $t - t^2/(2t) = t/2$. Subtracting $s/2$ removes the conditional mean. Translates of the same one-dimensional probability measure have W_1 equal to the translation distance, proving both ratios. □

The coefficient $1/2$ is not a failure of mixing: it is deterministic transport of retained information. Centering separates this transport from the fluctuation that is actually eliminated. On a longer Gaussian chain, analogous interpolation of retained endpoint data can make a full-coordinate Dobrushin row sum reach one even when the pinned fluctuation field is well controlled. Thus full-mode contraction is not representation invariant.

For the compact bridge, winding sectors make $\widetilde{\gamma}_t^s$ weakly s-dependent, but this dependence is exponentially suppressed in a small-field chart as $t \downarrow 0$. Section 9 verifies this numerically without promoting the observation to a four-dimensional theorem.

4 Fibered representation packages

Definition 4.1 (Fibered RG step)**.** *A fibered RG step is a tuple*

$$\mathfrak{R}_k = (X_k, \mu_k, X_{k+1}, \pi_k, Q_k, \mathcal{O}_k, \mathcal{G}_k, \mathcal{B}_k)$$

with:

1. *a probability space (X_k, μ_k) and a coarse space X_{k+1};*
2. *a measurable retained-data map $\pi_k : X_k \to X_{k+1}$ and $\mu_{k+1} = (\pi_k)_* \mu_k$;*
3. *a disintegration $\mu_k(dx) = \int Q_k(y, dx)\, \mu_{k+1}(dy)$ supported on $\pi_k^{-1}(y)$;*

4. *a gauge-invariant observable core $\mathcal{O}_k$ and an explicit transport rule $P_kF(y) = \int F(x)Q_k(y, dx)$;*

5. *a good event $\mathcal{G}_k$ on which conditional fiber influence is bounded by K_k, and a defect family $\mathcal{B}_k$ whose connected contribution is bounded by R_k.*

The retained base y is transported. The numbers K_k and R_k measure only loss of memory through the conditional fiber and defects, respectively.

For gauge theory, X_k may be a regulated connection space modulo specified local gauge actions; y must consist of explicitly chosen coarse covariant or invariant data. This definition does not solve gauge fixing: existence, measurability, locality, stabilizers, Gribov-type ambiguity, topology, and reflection compatibility are part of the open construction of $\mathfrak{R}_k$.

5 Exact covariance bookkeeping across scales

Let $\mathcal{F}_{k+1} \subseteq \mathcal{F}_k$ be the sigma-algebra generated by the retained map at step k, and write $P_kF = \mathbb{E}[F \mid \mathcal{F}_{k+1}]$.

Theorem 5.1 (One-step fiber decomposition, PROVED)**.** *For $F, G \in L^2(\mu_k)$,*

$$\mathrm{Cov}_{\mu_k}(F, G) = \mathrm{Cov}_{\mu_{k+1}}(P_kF, P_kG) + \mathbb{E}_{\mu_{k+1}}\left[\mathrm{Cov}_{Q_k(y,\cdot)}(F, G)\right]. \tag{2}$$

Proof. Insert $F = P_kF + (F - P_kF)$ and similarly for G. The mixed terms vanish after conditioning because $\mathbb{E}[F - P_kF \mid \mathcal{F}_{k+1}] = 0$. The covariance of the conditional means is the first term; the conditional covariance of the residuals is the second. □

Corollary 5.2 (Iterated fiber decomposition, PROVED)**.** *For a tower $\mathcal{F}_0 \supseteq \cdots \supseteq \mathcal{F}_n$, define $F_k = \mathbb{E}[F \mid \mathcal{F}_k]$ and $G_k = \mathbb{E}[G \mid \mathcal{F}_k]$. Then*

$$\mathrm{Cov}(F, G) = \mathrm{Cov}(F_n, G_n) + \sum_{k=0}^{n-1} \mathbb{E}[\mathrm{Cov}(F_k, G_k \mid \mathcal{F}_{k+1})]. \tag{3}$$

Proof. Apply Theorem 5.1 to (F_k, G_k) and telescope. □

Equation (3) is the precise form of representational dynamics used here. Each step declares what information survives in (F_{k+1}, G_{k+1}) and accounts separately for what is forgotten in the fiber covariance. It is an identity, not a decay estimate.

6 A conditional fibered rigidity theorem

We now state an abstract implication that makes the missing estimates visible. Let $d_k(A, B)$ denote separation of transported supports at scale k, and let $b > 1$ be the block factor. The theorem is deliberately formulated as a certificate: an application must provide all constants and verify all hypotheses without using clustering itself.

Theorem 6.1 (Fibered two-tier clustering certificate, CONDITIONAL)**.** *Suppose a tower of n fibered RG steps has the following properties for centered local observables F, G with initial support distance d.*

1. ***Quasilocal transport.*** *The transported observables F_k, G_k have norms at most A_F, A_G and effective separation $d_k \geq c_0 d/b^k - c_1$.*

2. ***Good-fiber loss.*** *For constants $C_f, m_f > 0$ and $0 \le K_k < 1$,*

$$|\mathbb{E}\,\mathrm{Cov}(F_k, G_k \mid \mathcal{F}_{k+1}); \mathcal{G}_k| \le C_f A_F A_G K_k e^{-m_f d_k}.$$

3. ***Defect channel.*** *Connected bad objects joining the two transported supports contribute at most*

$$C_b A_F A_G R_k e^{-m_b d_k},$$

where $R_k \ge 0$ is established by a dependent polymer or domination estimate.

4. ***Terminal base rigidity.*** *For some $C_*, m_* > 0$,*

$$|\,\mathrm{Cov}(F_n, G_n)| \le C_* A_F A_G e^{-m_* d_n}.$$

Then

$$|\,\mathrm{Cov}(F, G)| \le A_F A_G \left[C_* e^{-m_* d_n} + \sum_{k=0}^{n-1} (C_f K_k e^{-m_f d_k} + C_b R_k e^{-m_b d_k}) \right]. \tag{4}$$

In particular, any cutoff-uniform choice of $n = n(d)$ for which the right-hand side is bounded by Ce^{-md} proves exponential clustering on the specified observable core.

Proof. Take absolute values in (3). Split each conditional fiber term into its good and bad contributions and apply hypotheses 2 and 3. Apply hypothesis 4 to the terminal covariance and sum. The last sentence is exactly the stated uniform bound. □

The theorem does not say that $K_k + R_k < 1$ alone automatically gives physical exponential decay; scale conversion, support growth, amplitudes, and the number of steps matter. Its value is logical: retained information appears only in the transported terminal term, while every eliminated degree of freedom appears in a fiber increment. No relevant mode is incorrectly charged as a mixing failure.

7 A restricted connected-defect theorem

The simplest calibration makes bad blocks independent. It cannot describe four-dimensional gauge defects, but it gives a falsifiable numerical target.

Theorem 7.1 (Independent rooted defect clusters, PROVED)**.** *Let Γ be a graph of maximum degree Δ, and mark vertices bad independently with probability p. For a root o and $n \ge 1$,*

$$\Pr\{\textit{there exists a connected all-bad } A \ni o, |A| = n\} \le \min\{1,\ p(\Delta^2 p)^{n-1}\}.$$

Thus $\Delta^2 p < 1$ gives exponential decay in n for this event.

Proof. Every finite connected $A \ni o$ has a spanning tree. A depth-first traversal of that tree starting at o has at most $2(n-1)$ directed edge steps. There are at most $\Delta^{2(n-1)}$ such walks, hence at most that many encoded connected sets (overcounting is harmless). A fixed set of n distinct vertices is all bad with probability p^n. The union bound gives $\Delta^{2(n-1)} p^n = p(\Delta^2 p)^{n-1}$. □

In the Gaussian bridge of Theorem 3.1, declaring the centered fiber bad when $|R| > \rho\sqrt{t/2}$ gives

$$p_\rho = \Pr\{|N(0,1)| > \rho\} = \mathrm{erfc}(\rho/\sqrt{2}).$$

At $\rho = 3$, $p_3 \approx 0.0026998$. With $\Delta = 8$, $\Delta^2 p_3 \approx 0.1728 < 1$. This proves subcriticality only for the independent Gaussian calibration. In Yang–Mills theory, bad blocks are dependent and may contain large curvature, cut-locus effects, exceptional gauge orbits, or topological flux. A valid R_k must control those mechanisms rather than assume them away.

8 Regulator comparison at block level

Theorem 8.1 (Finite-product TV comparison, PROVED)**.** *For probability measures μ, ν and integer $m \geq 1$,*

$$\mathrm{TV}(\mu^{\otimes m}, \nu^{\otimes m}) \leq 1 - (1 - \mathrm{TV}(\mu, \nu))^m \leq m\ \mathrm{TV}(\mu, \nu).$$

Proof. Choose a maximal coupling of one μ and one ν sample, whose mismatch probability is $\mathrm{TV}(\mu, \nu)$. Couple m pairs independently. The product samples agree unless at least one pair mismatches, which has probability $1 - (1 - \mathrm{TV})^m$. Total variation is no larger than the mismatch probability of any coupling, and Bernoulli's inequality gives the second bound. □

Consequently convergence of heat-kernel and Wilson one-plaquette laws implies agreement for each fixed finite product block. It does *not* imply universality for a fixed physical block containing $m(a) \to \infty$ plaquettes: the displayed bound would require the one-plaquette discrepancy to be $o(1/m(a))$, unless correlations or a sharper comparison provide compensation. The same warning applies more strongly to interacting nonabelian blocks.

9 Executable compact calibration

The accompanying program evaluates the compact bridge on a periodic grid, using the median-corrected cumulative-distribution formula for geodesic W_1 on the circle. Boundary increments are $0.1\sqrt{t}$; defects use $\rho = 3$. The Wilson one-plaquette density is matched to the heat-kernel quadratic small-angle scale before TV comparison.

t	δs	full $W_1/\delta s$	fiber $W_1/\delta s$	p_{bad}	$64p_{\mathrm{bad}}$	one TV	16 TV bound
0.50	0.07071	0.498316	1.8×10^{-8}	0.002696	0.172523	0.063595	1.000000
0.20	0.04472	0.499999	$< 10^{-9}$	0.002697	0.172636	0.019918	0.318686
0.10	0.03162	0.500000	$< 10^{-9}$	0.002688	0.172036	0.009232	0.147705
0.05	0.02236	0.500000	$< 10^{-9}$	0.002719	0.174015	0.004477	0.071629

Eight automated tests verify normalization, the 1/2 tangent response, small centered response, the Gaussian-tail limit, defect subcriticality in the stated model, TV tensorization, and failure-closed input checks. These computations are reproducible evidence about the selected finite model, not computer-assisted proofs for nonabelian four-dimensional Yang–Mills theory.

10 The theorem dependency chain

The representational programme now factors the Clay target as follows:

$$\begin{aligned}&\text{regulated gauge measure} \xrightarrow{\text{gauge-covariant block/disintegration}} \text{base} + \text{conditional fiber}\\ &\xrightarrow{K_k \text{ and } R_k \text{ bounds}} \text{uniform fibered basin} \xrightarrow{\text{Theorem 6.1}} \text{Euclidean clustering}\\ &\xrightarrow{\text{continuum coherence + OS axioms}} \text{physical Hilbert space} \xrightarrow{\text{spectral transfer}} \mathrm{spec}(H) \subset \{0\} \cup [\Delta, \infty).\end{aligned} \tag{5}$$

Theorems 3.1–8.1 clarify the middle language but do not supply the four-dimensional arrows in (5). In particular, the existence of a cutoff sequence and compactness alone are insufficient for the stated endpoint if nontriviality, axioms, and a physical-unit gap are not preserved [1].

11 Frontier audit: what remains open

ID	Required result	Status
O1	Construct local, gauge-covariant four-dimensional block maps and disintegrations; specify retained loop data, fiber quotients, stabilizers, overlap consistency, locality, and reflection compatibility.	OPEN
O2	On good blocks prove a quotient Wasserstein or related influence bound $K_k < 1$, uniform in volume and cutoff, after retained modes are removed.	OPEN
O3	Control dependent large-field, exceptional-orbit, cut-locus, and topological defects by a connected polymer estimate with summable R_k.	OPEN
O4	Prove that the asymptotically free ultraviolet family enters the fibered basin with constants in physical units, without defining the basin by the desired clustering conclusion.	OPEN
O5	Prove convergence and regulator independence on a separating gauge-invariant observable core, together with reflection positivity, Euclidean covariance, symmetry, regularity, and nontriviality.	OPEN
O6	Reconstruct the Hilbert space and show that uniform Euclidean decay gives $\mathrm{spec}(H) \subset \{0\} \cup [\Delta, \infty)$ with $\Delta > 0$.	OPEN

The frontier theorem is O4, but it should not be attacked before O1–O3 are made quantitative. The immediate next nonabelian experiment is the smallest $SU(2)$ block with explicitly described retained conjugacy/loop data. It should be tested against noncommuting boundary changes and four adversarial defect classes: large curvature, holonomy near the cut locus, reducible stabilizers, and nontrivial flux. A positive numerical coefficient is useful only after the metric, quotient, and locality constants are fixed independently of the target claim.

12 Conclusion

The new ontology changes the proof obligation in a concrete way. An RG step is not a map that should erase all information. It is a typed transformation that *preserves* selected relational information in its base and *forgets* conditional fiber information with controlled error. The exact covariance decomposition records both operations. The mass-gap programme then asks whether infrared rigidity is stable across this chain and survives the continuum/reconstruction interface.

The two-plaquette bridge supplies a sharp sanity check: failure of full-mode contraction can coexist with perfect fiber contraction. This rescues a viable local target but does not establish it for four-dimensional Yang–Mills. The remaining mathematical core is now narrower and more honest: construct the nonabelian conditional fibers, prove their good-field contraction, control the dependent defect channel, and carry the resulting physical-unit estimate through a coherent continuum limit.

References

[1] A. Jaffe and E. Witten, *Quantum Yang–Mills Theory*, Clay Mathematics Institute Millennium Problem description (2000), https://www.claymath.org/wp-content/uploads/2022/06/yangmills.pdf.

[2] K. Osterwalder and R. Schrader, "Axioms for Euclidean Green's functions," *Communications in Mathematical Physics* **31** (1973), 83–112.

[3] K. Osterwalder and R. Schrader, "Axioms for Euclidean Green's functions II," *Communications in Mathematical Physics* **42** (1975), 281–305.

[4] B. K. Driver, "Two dimensional Yang–Mills theory: expectations for Wilson loops," heat-kernel/Wilson-loop notes, https://www.math.ucsd.edu/~bdriver/DRIVER/Papers/Drivers_Papers/A3-YM2-Expectations.pdf.